

EDCR: Evidence-Driven Chained Reasoning for Unified Traffic Anomaly Understanding

Duy Truong¹, Duy Khanh Dinh¹, Hoang Minh Hoang¹, Tien Cuong Nguyen¹,
Thanh Ha Cao¹, and Thien Van Luong^{2*}

¹ VNPT AI, VNPT Group, Vietnam

² Business AI Lab, College of Technology, National Economics University, Vietnam

Abstract. Traffic Anomaly Reasoning demands moving beyond simple binary anomaly detection to provide comprehensive, actionable diagnostics, requiring models to process heterogeneous queries ranging from exact-match verification to open-ended causal narratives. Recent benchmarks, such as Track 3 of the AI City Challenge, evaluate this paradigm by mandating a single foundational vision-language model to handle all diagnostic tasks on identical traffic clips. However, a fundamental challenge in this unified setup is the verification-generation gap: a single tuned backbone exhibits heavily asymmetric behavior, reliably making calibrated closed-form judgments while simultaneously struggling with hallucinations and semantic drift during extended narrative generation. Rather than attempting to train away this inherent structural conflict, we propose an **Evidence-Driven Chained Reasoning** (EDCR) approach that actively exploits it at inference. Our method centers on the **Cross-Question Evidence Sheet**, a compact block of grounded facts dynamically mined from the backbone’s own high-confidence closed-form decisions. By routing information from the reliable verifier to the uncalibrated generator, we condition the narrative generation zero-shot on this evidence sheet and refine it through candidate selection optimized for the target metric. Our proposed pipeline effectively bridges the asymmetry between the verifier and the generator, lifting the tuned backbone to achieve a 0.6669 mean score on the testing data and securing a Top-3 ranking on the official AI City Challenge Track 3 Leaderboard. All code implementations: github.com/AICity26-Track3-TAR.

Keywords: Video anomaly reasoning · Evidence-driven generation

1 Introduction

Roadside and dash-mounted cameras have made video evidence of collisions and near-misses routinely available to traffic operators [23], but turning it into actionable understanding requires more than a binary anomaly flag. Classical video anomaly detection reduces the problem to temporal localization or binary classification [25], whereas traffic management needs diagnostics: which agents

* Corresponding author. Email: thienlv@neu.edu.vn

were involved, what triggered the event, when it happened, and why. This has produced video-language benchmarks posing causal, counterfactual and descriptive queries over anomaly footage [6, 7, 13, 26, 34].

The Traffic Anomaly Reasoning (TAR) benchmark consolidates a highly heterogeneous spectrum of tasks onto identical video clips. These tasks range from strict closed-form verification to complex open-ended causal linkage. Closed-form tasks, such as Binary and Multiple-Choice Questions, require exact token matches evaluated via rigid regex extraction. Conversely, causal linkage tasks demand long-form, text-explainable justifications for anomalous events. Recent evaluations, such as Track 3 of the 10th AI City Challenge [27], mandate a single foundational model to solve all these query types simultaneously, creating a direct architectural conflict. Exact-match verification requires calibrated, low-entropy probability distributions over a single token. Conversely, the seven free-form narrative types, which dominate the leaderboard score, demand high-recall, expressive text evaluated by BERTScore against human references. Because specialized task-specific heads are prohibited, conventional multi-task training inevitably trades one objective against the other [17].

Our starting point is that a single tuned backbone does not balance this conflict; it heavily favors one side. When functioning as a *verifier* for closed-form queries, the model yields well-calibrated, high-confidence judgments. Conversely, when acting as a *generator* for open-ended narratives, the same weights produce less reliable text susceptible to unverified statements and semantic drift. We define this discrepancy as the verification-generation gap, reflecting a video-domain instance of the asymmetry between the verifier and the generator noted in text-only settings [5, 8]. Attempting to train away this gap risks catastrophic forgetting of the robust verification capabilities we require. Therefore, we actively exploit it. The core principle of our architecture relies on a strict unidirectional flow of information: *every mechanism below moves information from the verifier to the generator, and none the other way*.

We tune a single Qwen3-VL-32B [2] backbone once with LoRA [11], freezing the vision encoder and projector, under one strict discipline: each task is bound to a fixed output format. By restricting the loss for closed-form verification tasks to the leading token, we ensure the first-token distribution acts as a calibrated verification judgment rather than the start of a free-form rationale. At inference, the merged checkpoint enters a deterministic chain (Figure 2) built around the **Cross-Question Evidence Sheet**: a compact block of natural-language facts mined directly from the backbone’s own high-confidence closed-form verification decisions regarding the same clip. Open-ended narrative generation is subsequently conditioned on this sheet and re-ranked by a video-grounded Chain-of-Verification [5] and Minimum Bayes Risk decoding [8, 14] under the benchmark’s official grader.

We call the system Evidence-Driven Chained Reasoning (**EDCR**). It is *evidence-driven* because every narrative is conditioned on grounded facts rather than on the clip alone, and *chained* because those facts are produced and consumed by an ordered sequence of reasoning components.

In summary, our contributions are twofold:

- **Evidence-Driven Chained Reasoning (EDCR):** We introduce an inference pipeline that explicitly exploits the verification-generation gap in VLMs. By aggregating the model’s calibrated closed-form judgments into a **Cross-Question Evidence Sheet**, we route extracted facts to condition open-ended generation, drastically suppressing unverified narratives.
- **Empirical Validation and Rigorous Ablation:** EDCR achieves a 0.6669 mean score and a Top-3 ranking on the official AI City Challenge Track 3 Leaderboard. Through a component-by-component ablation detailing both positive and negative results, we isolate the exact impact of each mechanism, providing a generalizable framework for reasoning across diverse tasks with divergent evaluation metrics.

2 Related Work

Video anomaly reasoning and the AI City Challenge. Early video anomaly analysis framed the task as weakly-supervised temporal detection [25]. The field has shifted to Video Anomaly Reasoning (VAR) [9, 12], asking models to explain the *what, why, and how* of events [6, 7], including open-world scenarios [26], hierarchical temporal analysis [34] and context-aware anomalies under unified generative evaluation [33]. In intelligent transportation, AccidentBench [10] and WTS [13] report gaps in pedestrian-centric spatio-temporal reasoning. The AI City Challenge [27, 28, 31] follows this trend: its Track3 introduces TAR, requiring one foundational model to solve ten heterogeneous tasks from multiple-choice verification to open-ended causal linkage on identical clips. Our method targets the resulting tension between classification and generation.

Vision-language backbones and parameter-efficient adaptation. Early vision-language models [1, 15, 22] were extended to video by sampling a few frames and pooling their features over time [16, 19]. This design loses the fine spatial and temporal details that anomaly reasoning relies on. The Qwen-VL lineage advanced continuous temporal modeling: Qwen2-VL’s dynamic resolution [30], Qwen2.5-VL’s absolute time encoding [3], and Qwen3-VL [2], which integrates Interleaved-MRoPE and DeepStack fusion for text-timestamp alignment. We use one Qwen3-VL-32B backbone for all tasks, with LoRA [11] on the language decoder and frozen visual representations.

Test-time reasoning and multi-stage inference. Recent methods use multi-stage inference and test-time scaling, extracting intermediate evidence before synthesis. Chaining sub-tasks yields factual priors that refine narratives without additional fine-tuning [17], and LATERN [21] applies test-time context-aware principles to long-form videos.

Verification and risk-minimizing decoding. Self-critique pipelines build on a simple observation: a model is more reliable when it judges an answer than

when it writes one. For example, Chain-of Verification [5] splits a draft into atomic claims and asks the model to check each claim separately. Minimum Bayes Risk (MBR) decoding [8, 14] replaces mode-seeking search with selection under an explicit utility, and is effective exactly when sequence probability is misaligned with the metric. Our proposed EDCR combines both and differs on two points: our verification signal is a calibrated first-token readout rather than free-text self-critique, so it inherits the calibration our tuning enforces; and our MBR utility is the official grader itself. Position bias in multiple-choice scoring is well documented [20, 29, 35]; we remove it with a rotation-based scoring step described in Section 4.3.

3 Motivation

Before detailing our reasoning architecture, we analyze the constraints of the evaluation metric and conduct a controlled pilot study to expose the inherent behavioral discrepancies of the tuned vision-language backbone. These observations establish the empirical foundation for our proposed method.

3.1 Task and Metric Analysis

Task and data. TAR [27] requires evaluating ten task types per clip. These include two decision types, namely BCQ and MCQ, along with their open-ended variants requiring text justifications, specifically BCQ-OE and MCQ-OE; five explanation types (openQA, causal linkage, scene description, temporal description, summarization); and temporal localization. The training split contains 44,040 pseudo-labeled items across 3,670 CCTV videos (965 anomalous, 2,705 normal; 26.1 hours) from eight public sources [4, 10, 12, 18, 24, 25, 32, 36]. These annotations, including chain-of-thought traces, were generated via a VLM pipeline, utilizing existing NVIDIA human annotations as supplementary context for 910 of the clips. The test split differs on exactly those axes: 960 human-verified items over 80 clips trimmed from 17 public YouTube crash and dashcam videos. The gap is one of style as much as content, so reproducing the training label distribution underperforms on the human-curated references.

The metric drives the design. Exact-match accuracy scores BCQ and MCQ; BERTScore scores every free-form type. The leaderboard number is the unweighted mean of the nine graded types; temporal-localization mIoU is excluded. Three consequences shape everything below. *(i)* The seven open-ended narrative tasks account for roughly 78% (six out of nine metrics) of the final leaderboard score. In these tasks, matching the expected length and linguistic style of the reference answers impacts the BERTScore more heavily than minor improvements in visual recognition. Therefore, addressing the divergent nature of the evaluation metric requires prioritizing these narrative types. *(ii)* MCQ and MCQ-OE share one decision, so a single calibrated letter serves two columns. *(iii)* Temporal localization earns no mean points and receives no learned investment; it is retained only as the source of the time-window fact the narratives reuse.

Structural couplings. The test *questions* reuse the training templates verbatim, while the test *answers* are human-curated. This exposes label-free regularities that the reasoning chain exploits deterministically. First, timestamps in temporal-description and causal-linkage questions often duplicate the clip’s localization window. Second, 3,666 out of 3,669 training videos feature exactly one *Yes* and one *No* in their binary question pairs. Third, 72 of the 80 MCQ stems reappear identically as MCQ-OE stems. Finally, entities such as agents, scene attributes, and candidate root causes are explicitly provided within the questions themselves. None requires test labels. We distinguish throughout between mechanisms that exploit these couplings and those that do not, discussed in Section 5.2.

3.2 Verification-Generation Gap

A critical challenge in unified modeling is the gap between exact match tasks and narrative tasks. The model predicts single tokens for yes or no questions with high reliability. However, it struggles with semantic drift during long text generation. We define this discrepancy as the inherent behavioral asymmetry of the backbone. To quantify this, we conduct a controlled analysis on a held-out split of 742 validation items using a single fine-tuned Qwen3-VL-32B checkpoint [2]. Because the ground-truth references for this split are pseudo-labeled, we focus on relative performance deltas (Δ) rather than absolute scores, treating any variance below 0.005 as statistical noise.

On the 149 closed-form queries, the verification operator P_θ , which predicts the first token, demonstrates strong calibration as shown in Figure 1. It achieves an Expected Calibration Error, or ECE, of 0.021 and an overall accuracy of 0.96. Notably, 88% of its predictions exceed a confidence threshold of 0.9, where accuracy reaches 1.00. Furthermore, predictive accuracy scales monotonically with the confidence margin m_q , which is the probability difference between the top two tokens, peaking at absolute certainty when $m_q \geq 0.8$.

Conversely, the exact same backbone weights struggle with open-ended generation, producing narratives with modest BERTScores ranging from 0.41 to 0.52 (Table 1, top row). This confirms that the model is significantly more reliable at closed-form verification than at open-ended narrative generation. A naive approach might employ self-verification to correct these narratives; however, when tasked with verifying its own generated sentences, P_θ erroneously endorses 98% of them, as both operators share the same underlying perceptual limitations on difficult clips.

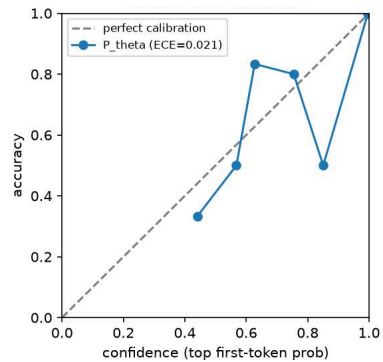


Fig. 1: Calibration of the Verification Probe. The per-bin accuracy of the closed-form operator P_θ closely tracks its confidence, yielding ECE of 0.021. The majority of decisions exhibit high confidence and near-perfect accuracy.

Table 1: Impact of Contextual Grounding. Evaluation on the held-out split. Every row is a greedy render sharing one template, length budget, and grader; only the provided context changes. Δ is measured against the empty template. **Ours** indicates the dynamically predicted sheet, while *Oracle* serves as an upper bound built from ground-truth references. The bottom block ablates individual fields from our predicted context.

Context Variant	B-OE	QA	CL	SD	TD	Summ	Mean	Δ
Baseline (w/o context)	0.6634	0.4922	0.5006	0.5239	0.4056	0.4672	0.6319	-0.0010
Empty template	0.6575	0.5023	0.4995	0.5198	0.4134	0.4696	0.6329	0.0000
Mismatched context	0.6095	0.4707	0.3919	0.4928	0.2896	0.4291	0.5909	-0.0420
Ours (Predicted context)	0.6526	0.5036	0.5306	0.5190	0.4165	0.5140	0.6412	+0.0083
Oracle (GT context)	0.7444	0.5032	0.5388	0.5309	0.4176	0.5228	0.6547	+0.0218
<i>Ablation of predicted context fields</i>								
w/o Observations	0.6577	0.4994	0.5281	0.5210	0.3867	0.5076	0.6372	-0.0040
w/o Consequence	0.6541	0.5013	0.5289	0.5082	0.4123	0.5051	0.6383	-0.0029
w/o Root cause	0.6514	0.5071	0.5287	0.5193	0.4159	0.5058	0.6403	-0.0009
w/o Agents	0.6515	0.5059	0.5333	0.5201	0.4208	0.4984	0.6405	-0.0007
w/o Scene	0.6582	0.5137	0.5328	0.5295	0.4088	0.5127	0.6433	+0.0021

3.3 Pilot Study

To verify whether explicitly routing verified facts to condition the generation process can resolve this gap, we isolate the impact of contextual conditioning in Table 1 by fixing all hyperparameters and varying only the provided context. Providing an *Empty template* yields identical performance to the raw baseline, indicating that structural formatting alone provides no intrinsic benefit. Conversely, supplying a *Mismatched context* (correct format, incorrect video facts) degrades the score by -0.0420 relative to the empty baseline. This confirms that the model actively utilizes the semantic *content* of the sheet rather than merely mimicking its structural scaffold, meaning incorrect priors heavily penalize generation.

Most importantly, our **Predicted context** (populated dynamically by the model’s own calibrated P_θ predictions) improves the mean score by $+0.0083$. This gain is highly concentrated in narrative tasks where the sheet explicitly supplies missing context. An *Oracle* context (constructed from the split’s reference answers) yields an additional $+0.0218$ improvement, highlighting the remaining performance headroom. Finally, ablating individual components reveals that the *Observations* field is the most crucial for performance. These empirical findings dictate our system design: we must establish a unidirectional reasoning chain that automatically constructs this evidence sheet at inference time, while mitigating the propagation of occasional incorrect priors.

4 Proposed Method

Motivated by the empirical observations in Section 3, we design our system as a two-phase chained reasoning pipeline. First, we apply Supervised Fine-Tuning

(SFT) purely to standardize the output format of the backbone for each task. Second, we introduce a deterministic reasoning architecture to iteratively optimize the factual content of the responses without further altering the model weights. This pipeline addresses both closed-form queries and open-ended narratives (Figure 2).

4.1 Notation

Let x denote a clip, $\mathcal{Q}(x)$ for its set of questions, $q \in \mathcal{Q}(x)$ for one question, $i = (x, q)$ for a graded item, and $\tau(q)$ for its task type, split into closed-form $\mathcal{T}_{\text{cls}} = \{\text{BCQ}, \text{MCQ}\}$ and narrative $\mathcal{T}_{\text{nar}}$ types. Let θ denote the parameters of the backbone model, and u denote the text template derived from query q . We use the single tuned backbone f_θ in two distinct modes and never mix them: a *Verification Operator* $P_\theta(\cdot | x, u)$ returning the first-token distribution over an answer alphabet, and a *Generation Operator* $G_\theta(x, u; \sigma)$ returning a decoded string under decoding regime σ (greedy or sampling).

The *verification-generation gap* lies in the observation that P_θ is calibrated while G_θ is not reliably grounded. Therefore, EDCR routes information strictly from P_θ to G_θ and never the reverse. Every reasoning component acts as a filter or refinement step. It overrides a previous prediction only when a specific confidence gate is triggered. Otherwise, the prediction passes through unchanged. Consequently, the reasoning chain is designed to refine the input, mitigating the risk of degradation. All remaining symbols are defined where they are first used, with numeric values collected in Section 5.1.

4.2 Model Tuning Strategy

We optimize a single Qwen3-VL-32B backbone [2] via SFT. Our primary objective during this phase is to constrain the model to produce a strict, fixed answer format for each query type. To achieve parameter-efficient adaptation, we apply Low-Rank Adaptation (LoRA) [11] exclusively to the language decoder. Specifically, for a frozen weight matrix W_0 , we learn the weight update $W = W_0 + \frac{\alpha}{r} \mathbf{B}\mathbf{A}$, where $\mathbf{B} \in \mathbb{R}^{d_{\text{out}} \times r}$ and $\mathbf{A} \in \mathbb{R}^{r \times d_{\text{in}}}$ with rank $r \ll \min(d_{\text{out}}, d_{\text{in}})$. This configuration updates approximately 0.40% of the backbone parameters. Crucially, we maintain a frozen vision encoder and cross-modal projector; this prevents the perceptual features from drifting toward the label noise inherent in the auto-generated supervision. During SFT, we minimize the standard causal language-modeling loss exclusively over the response tokens.

Strict format regularization. Each task type is bound to a single output format (e.g., a bare *Yes/No*, a single option letter, a leading decision token followed by one sentence, or a full narrative paragraph). For closed-form verification tasks, the target is restricted to a single generated token representing the final decision. By constraining the target to a single token, the first-token distribution acts as a calibrated judgment. This prevents the model from generating an open-ended rationale that it might later contradict. Consequently, we can utilize first-token

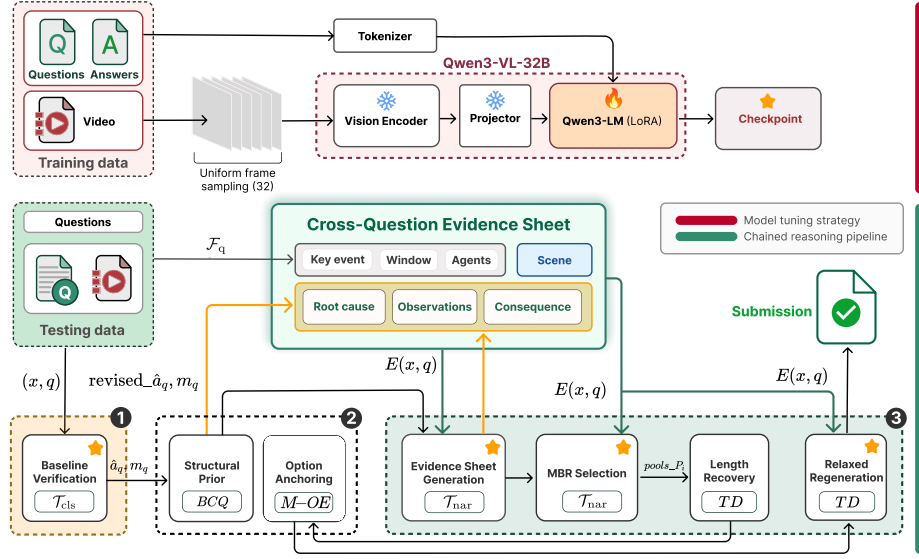


Fig. 2: Architecture of Evidence-Driven Chained Reasoning (EDCR). In the model tuning strategy (top red), a single Qwen3-VL-32B [2] is fine-tuned using LoRA [11] to standardize task-specific output formats. The reasoning chain pipeline (bottom green) flows through three cohesive blocks: *Fact Extraction* ① via the Verification Operator P_θ , *Conditioned Generation* ② via the Generation Operator G_θ , and *Benchmark-Specific Alignment* ③. Yellow stars indicate operations requiring forward passes through the model.

log-probability as a reliable Verification Operator (P_θ) during inference. This format discipline also ensures that every subsequent component regenerates answers in the exact format observed during SFT. The only prompt element unseen during tuning is the **Cross-Question Evidence Sheet**. Because it is structured as natural-language declarative statements rather than a novel syntax, the fine-tuned Qwen3-VL-32B backbone integrates this context in a training-free manner.

4.3 The Evidence-Driven Reasoning Architecture

At inference, the tuned model generates final predictions through a structured reasoning architecture, detailed in Table 2. To distinguish our core methodology from competition-specific adaptations, we conceptually divide this pipeline into three components. The transferable reasoning framework strictly enforces a unidirectional information flow from the verification operator P_θ to the generation operator G_θ through two core modules: *Calibrated Fact Extraction* and *Evidence-Conditioned Generation*. Subsequently, *Benchmark-Specific Alignment* optimizes the inputs and outputs for the official evaluation metrics.

Table 2: Dataflow of the EDCR Architecture. Input, output, and model calls for each sequentially executed component. Components are grouped into the three core reasoning blocks. [†] Denotes benchmark-specific alignments.

Component	Consumes	Produces	Task	Call
Fact Extraction via P_θ	x, q	$\hat{a}_q, m_q$	$\mathcal{T}_{\text{cls}}$	P_θ
Structural Prior [†]	$\hat{a}_q, m_q$	Revised $\hat{a}_q$	BCQ	–
Conditioned Generation via G_θ	$\hat{a}_q, m_q \Rightarrow E(x, q)$	E ; Draft narratives	$\mathcal{T}_{\text{nar}}$	P_θ, G_θ
MBR Selection	$E(x, q)$, Draft narratives	pool $\mathcal{P}_i$; Selected narrative	$\mathcal{T}_{\text{nar}}$	P_θ, G_θ
Length Recovery	$\mathcal{P}_i$, Selected narrative	Extended narrative	TD	–
Option Anchoring [†]	$\hat{a}_q, m_q$	Anchored narrative	MCQ-OE	–
Relaxed Regeneration	$E(x, q)$, Extended narrative	Final TD narrative	TD	G_θ

Fact Extraction To supply the calibrated decisions $\hat{a}_q$ and confidence margins m_q consumed by the reasoning process, we employ a baseline verification (P_θ) mechanism. During fine-tuning, closed-form targets are strictly constrained to a single generated token. Thus, the first-token distribution directly yields a calibrated verification judgment.

Given the input video x and text template u for option ordering ρ , P_θ produces a logit $\ell_a^{(\rho)}$ for each valid token. We apply a softmax over the answer alphabet $\mathcal{A}_q$, consisting of ‘Yes, No’ for BCQ and the released letters for MCQ, to obtain normalized probabilities.

To eliminate the position bias inherent in language models when evaluating multiple-choice queries [20, 29, 35], we process R cyclic rotations $\pi_1, \dots, \pi_R$ of the option texts and average the resulting distributions:

$$\bar{p}(a) = \frac{1}{R} \sum_{\rho=1}^R p^{(\rho)}(\pi_\rho(a)). \quad (1)$$

Since MCQ and MCQ-OE tasks share identical option sets, their distributions are pooled. The final decision is determined as $\hat{a}_q = \arg \max_a \bar{p}(a)$. The margin $m_q = \bar{p}_{(1)} - \bar{p}_{(2)}$ explicitly quantifies the probability difference between the top two tokens, serving as the confidence threshold for subsequent reasoning components.

Evidence-Conditioned Generation Open-ended narrative tasks dominate the evaluation metric, yet the Generation Operator G_θ frequently produces unverified statements and semantic drift on these queries. Conversely, the Verification Operator P_θ reliably answers closed-form queries utilizing the same pre-trained weights. To mitigate this verification-generation discrepancy, we introduce the Cross-Question Evidence Sheet. This representation acts as a natural-language interface that conditions G_θ using factual priors extracted by P_θ . The sheet constitutes a partial map from the visual attributes $\mathcal{F} = \{\text{KeyEvent, Window, Agents, Scene, RootCause, Consequence, Observations}\}$ to declarative statements.

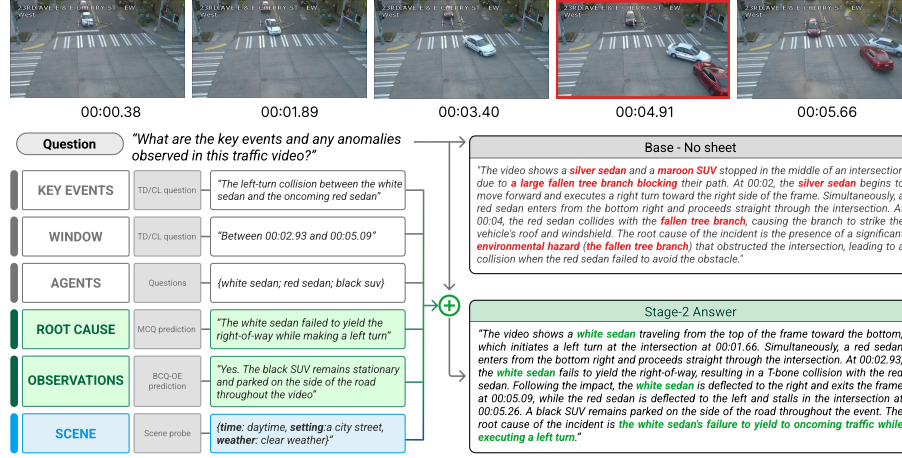


Fig. 3: Qualitative Example of the Evidence Sheet. A source-tagged sheet is assembled from the backbone’s own calibrated probes and injected at inference. Conditioning on it corrects the unconditioned model’s hallucinations (red: silver/maroon agents, fallen-tree-branch cause) into a factually grounded account (green: white-sedan failure-to-yield T-bone, impact at 00:04.91), with no change to the tuned weights.

Cross-Question Evidence Sheet

The question-derived attributes $\mathcal{F}_q = \{\text{KeyEvent}, \text{Window}, \text{Agents}\}$ are leveraged directly from the query set $\mathcal{Q}(x)$. These attributes capture the physical meaning of the traffic scene as illustrated in Figure 3. For instance, the Window attribute specifies the exact incident time. The Scene attribute describes the environmental conditions. The remaining cross-question attributes are assembled, when answering a narrative question q , from the closed-form decisions on the other questions about the same clip:

$$E(x, q)[\phi(q')] = \text{stmt}(q', \hat{a}_{q'}) \quad \text{for } q' \in \mathcal{Q}(x) \setminus \{q\} \text{ with } \tau(q') \in \mathcal{T}_{\text{cls}}, m_{q'} \geq \gamma, \quad (2)$$

where $\hat{a}_{q'}$ is the decision on q' and $m_{q'}$ is its confidence margin. The function ϕ maps a question to the corresponding attribute. The function stmt deterministically converts a discrete choice into a declarative statement. For example, this function maps a selected option A to its full declarative text stating the white sedan failed to yield. Finally, γ is the confidence threshold.

Equation 2 formulates the core conditioning mechanism via a risk-adaptive confidence gate. As demonstrated in Section 3.2, the Verification Operator P_θ inherently produces highly confident predictions. Therefore, we set the baseline threshold to $\gamma = 0$ for structurally constrained attributes to maximize factual recall. Conversely, for open-vocabulary attributes prone to hallucinations like the Scene attribute, we enforce a stricter high-confidence floor of $\gamma = 0.5$. This mechanism requires no additional training phases because the language decoder

integrates the assembled sheet zero-shot. After conditioning on this context, G_θ generates a raw narrative $\tilde{y}_i$. Because BERTScore heavily penalizes gross length mismatches, each generation is calibrated to the per-task median reference length $\beta_{\tau(q)}$ with tolerance δ , truncated at a sentence boundary to the longest prefix within budget:

$$\hat{y}_i^{(2)} = \tilde{y}_{i,1:s^*}, \quad s^* = \max\{s : \text{len}(\tilde{y}_{i,1:s}) \leq (1 + \delta) \beta_{\tau(q)}\}. \quad (3)$$

To refine the narrative outputs and mitigate the propagation of incorrect priors from the evidence sheet, we employ a candidate selection mechanism. For each narrative query, we sample a pool $\mathcal{P}_i$ of K candidate answers. To enforce decorrelation, one baseline candidate is generated using the complete evidence sheet, while the remaining $K-1$ candidates are generated by masking exactly one prediction-derived attribute. Each candidate c is evaluated using a dual-scoring formulation:

$$v(c) = \frac{1}{|\mathcal{C}(c)|} \sum_{z \in \mathcal{C}(c)} P_\theta(\text{Yes} \mid x, \text{T}_{\text{BCQ}}(z, \emptyset)); \mu(c) = \frac{1}{K-1} \sum_{c' \in \mathcal{P}_i \setminus \{c\}} \text{Sim}_{\text{off}}(c, c'),$$

$$\hat{y}_i^{(3)} = \arg \max_{c \in \mathcal{P}_i} [\mu(c) + \lambda v(c)]. \quad (4)$$

The first component is a Minimum-Bayes-Risk (MBR) utility $\mu(c)$ [8, 14], measuring consensus under the official grader’s metric Sim_{off} . The second component is a self-verification score $v(c)$ [5]. Given that P_θ and G_θ share the same underlying parameterization, a self-verification mechanism inherently struggles to correct fundamental perceptual blind spots. Consequently, the MBR utility $\mu(c)$ acts as the primary consensus filter to suppress uncorrelated hallucinations, whereas $v(c)$ serves as a secondary regularizer for optimal phrasing.

Finally, to address temporal descriptions that fall short of human references due to the initial length budget, we apply two transferable generative recovery steps. First, we select the longest candidate within the pool $\mathcal{P}_i$ that extends the chosen answer. Second, we exclusively re-run the generation and MBR selection for this task under a relaxed length budget, enabling G_θ to reach its inherent stopping point without over-generation.

Benchmark-Specific Alignment While the core architecture establishes a transferable framework for video reasoning, the evaluation metric and specific annotation protocols of the AI City [27] introduce strong structural biases. To bridge the gap between generalized reasoning and benchmark-specific expectations, we implement a set of deterministic refinements. Operationally, as detailed in Table 2, these refinements are deterministically interleaved with the core reasoning modules to satisfy metric constraints at specific points in the pipeline. Although these interventions yield substantial empirical gains, they explicitly exploit dataset construction biases and are structurally decoupled from the core reasoning architecture.

Structural Prior. The training distribution reveals that the two binary queries for a given clip overwhelmingly form a complementary pair. If P_θ outputs identical answers for a pair, we resolve the contradiction by retaining the decision with the higher margin m_q and flipping the less confident one. This logic is applied to the extracted decisions before they populate the evidence sheet.

Option Anchoring. For the MCQ-OE task, reference annotations typically restate the chosen option text verbatim rather than providing a paraphrased justification. Since open-ended generation naturally diverges from this constraint, we bypass G_θ entirely when $m_q \geq \gamma$. Instead, the final output is deterministically anchored by appending the normalized option text to the chosen letter.

5 Experiments

5.1 Implementation Details

Tuning configuration. We apply LoRA [11] ($r = 16, \alpha = 32$, bfloat16) to all linear projections of the language model, while keeping the vision encoder and projector frozen. The backbone is fine-tuned for two epochs using a peak learning rate of 2×10^{-4} with a cosine decay schedule, processing 32 uniformly sampled frames per clip under a 4096-token sequence limit. Finally, the LoRA weights are merged into the backbone to yield a single deployment checkpoint for all inference processing.

Inference and runtime. During the baseline verification (P_θ) phase, we evaluate $R = 4$ option rotations. The cross-question margin gate is set to $\gamma = 0$, admitting all parseable closed-form decisions into the **Cross-Question Evidence Sheet**, whereas scene probes require a stricter probability floor of $\gamma_{sc} = 0.5$. For the MBR selection, we sample a pool of $K = 8$ narratives at temperature 0.8, blending the scores ($\lambda = 0.3$) and extracting a maximum of eight atomic claims per candidate. The standard length tolerance $\delta = 0.3$ is relaxed to $\delta' = 0.8$ alongside an expanded pool $K' = 16$ specifically for the temporal description. To perfectly align with the official evaluation, Sim_{off} utilizes BERTScore-F1 (roberta-large); its $O(K^2)$ pairwise computation cost is efficiently amortized via global batching. Executed on four A100-80GB GPUs, the entire pipeline is deterministic under a fixed seed (save for candidate sampling), requiring ~ 32 hours for two SFT epochs and approximately 3 hours to process the full evaluation split.

5.2 Main Results

Table 3 details the sequential integration of each reasoning component in our pipeline. The cumulative increments confirm our design logic: each mechanism selectively improves its targeted task type without degrading others. Specifically, the integration of the *Evidence Sheet* yields the most substantial gain (+0.0346), as it is the sole component conditioning all open-ended narrative types simultaneously. *MBR selection* and the two *length recovery* mechanisms further refine these

Table 3: Cumulative Ablation Study. Sequential integration of each component on the evaluation split. Δ represents the cumulative mean increment. † denotes benchmark-specific alignments.

Component	BCQ	MCQ	B-OE	M-OE	QA	CL	SD	TD	Summ	Mean	Δ
Baseline (P_θ)	0.8938	0.9250	0.6124	0.8615	0.4368	0.4325	0.3879	0.3518	0.4351	0.5930	–
+ Structural Prior †	1.0000	0.9250	0.6124	0.8615	0.4368	0.4325	0.3879	0.3518	0.4351	0.6047	+0.0117
+ Evidence Sheet	1.0000	0.9250	0.6177	0.8615	0.4797	0.5096	0.4120	0.4272	0.5211	0.6393	+0.0346
+ MBR Selection	1.0000	0.9250	0.6337	0.8615	0.5012	0.5281	0.4368	0.4253	0.5251	0.6485	+0.0092
+ Length Recovery	1.0000	0.9250	0.6337	0.8615	0.5012	0.5281	0.4368	0.4747	0.5251	0.6540	+0.0055
+ Option Anchoring †	1.0000	0.9250	0.6337	0.9667	0.5012	0.5281	0.4368	0.4747	0.5251	0.6657	+0.0117
+ Relaxed Regeneration	1.0000	0.9250	0.6337	0.9667	0.5012	0.5281	0.4368	0.4853	0.5251	0.6669	+0.0012
<i>Transferable components</i>	+0.0505		<i>Benchmark-specific alignments†</i>				+0.0234		Total	+0.0739	

narratives (+0.0092, +0.0055, and +0.0012, respectively). Regarding closed-form queries, the *Structural prior* adjustment resolves BCQ inconsistencies (+0.0117), while *Option anchoring* exclusively lifts MCQ-OE performance from 0.8615 to 0.9667.

Transferable versus benchmark-specific gains. We categorize the total +0.0739 improvement by generalization capability. *Transferable reasoning components* rely strictly on the inherent properties of the backbone, contributing the majority of the gain (+0.0505). Conversely, *Benchmark-specific alignments* contribute +0.0234 (31.7% of the total). These alignments explicitly exploit dataset-specific label construction conventions (complementary BCQ pairs and the option-derived phrasing of the MCQ-OE references). While valid under the challenge protocol, these heuristics do not reflect generalized reasoning and are void on out-of-domain evaluation splits lacking such structural priors.

5.3 Discussion

Our empirical evaluation also reveals negative results and persistent failure modes, providing key insights into the boundaries of the verification-generation gap.

Negative results. Three mechanisms we expected to yield improvements did not. (i) *Failure of Generative Option Anchoring:* Routing the MCQ-OE task through the Evidence Sheet generation and Verify+MBR pipeline degraded the score by -0.0495 ($0.9236 \rightarrow 0.8741$). Where the reference answer strictly demands re-stating the option text, open-ended generation is inherently ineffective. Our deterministic Option Anchoring correctly circumvents this, restoring performance to 0.9300 (and ultimately 0.9667). (ii) *Redundancy of closed-form self-verification:* Applying the verification fusion directly to the *closed* MCQ letter produced a net-zero flip rate. The generator and the verifier share the same underlying perceptual weights; therefore, a same-model verifier inherently struggles to correct fundamental perceptual errors. Verification is only effective on free-text claims where errors decorrelate, never on closed-form letters. (iii) *Ceiling of length recovery:* For the Temporal Description (TD) task, relaxing the length budget tolerance δ' beyond 0.8 during the relaxed-budget regeneration resulted in a

0.0000 performance delta. This confirms that the model’s inherent temporal localization capability is the binding constraint, not the text generation budget.

Residual failures. Despite the robust architecture, three failure modes persist. First, regarding *unfiltered sheet hallucinations* during evidence aggregation, the model relies on soft prompting to resolve conflicting facts as the Evidence Sheet is consumed zero-shot. Ablation shows that while incorrect *root causes* are robustly ignored, hallucinated *agent* or *scene* attributes leak into the final narrative in 70% and 85% of cases, respectively. Second, this leads to *correlated candidate survival* during the MBR selection. Since all K candidates share the same Evidence Sheet, sheet-level hallucinations propagate across the entire pool $\mathcal{P}_i$. MBR selection (Eq. (4)) struggles to filter these out as it favors consensus, leaving the verification penalty $\lambda v(c)$ as the only, and sometimes insufficient, defense. Finally, *partial BCQ-OE inconsistency* emerges during structural prior alignment: flipping a low-confidence BCQ leading token (e.g., ‘Yes’ to ‘No’) for complementary pairing improves the metric but leaves the unedited reasoning clause locally inconsistent. Separately, MCQ errors remain unchanged (0.9250) as they stem from fundamental perceptual limits rather than formatting, placing them beyond the reach of our inference-time chain.

6 Conclusion

We presented Evidence-Driven Chained Reasoning (EDCR), an evidence-driven chained reasoning system for unified traffic anomaly understanding. By shifting the paradigm from joint multi-task optimization to test-time reasoning, our design explicitly exploits the verification-generation gap: while the tuned backbone serves as a highly reliable closed-form verifier, it remains susceptible to unverified statements during open-ended generation. To bridge this asymmetry, EDCR applies strict format regularization, transforming the first-token distribution into a calibrated verification probe. These judgments dynamically populate the **Cross-Question Evidence Sheet**, establishing an unidirectional flow of factual priors to condition the narrative generator. Integrated with *MBR selection*, this pipeline systematically suppresses semantic drift, achieving a 0.6669 mean score and securing a Top-3 ranking on the official Leaderboard. While the multi-component selection process increases inference latency, and dataset-specific alignments must be disabled for out-of-domain generalization, EDCR provides a robust blueprint for bridging the verification-generation gap in multimodal reasoning.

References

- [1] Alayrac, J.B., Donahue, J., Luc, P., Miech, A., Barr, I., Hasson, Y., Lenc, K., Mensch, A., Millican, K., Reynolds, M., et al.: Flamingo: A visual language model for few-shot learning. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2022)
- [2] Bai, S., Cai, Y., Chen, R., Chen, K., Chen, X., Cheng, Z., Deng, L., Ding, W., Gao, C., Ge, C., Ge, W., Guo, Z., Huang, Q., Huang, J., Huang, F., Hui, B., Jiang, S., Li, Z., Li, M., Li, M., Li, K., Lin, Z., Lin, J., Liu, X., Liu, J., Liu, C., Liu, Y., Liu, D., Liu, S., Lu, D., Luo, R., Lv, C., Men, R., Meng, L., Ren, X., Ren, X., Song, S., Sun, Y., Tang, J., Tu, J., Wan, J., Wang, P., Wang, P., Wang, Q., Wang, Y., Xie, T., Xu, Y., Xu, H., Xu, J., Yang, Z., Yang, M., Yang, J., Yang, A., Yu, B., Zhang, F., Zhang, H., Zhang, X., Zheng, B., Zhong, H., Zhou, J., Zhou, F., Zhou, J., Zhu, Y., Zhu, K.: Qwen3-vl technical report. arXiv preprint arXiv:2511.21631 (2025)
- [3] Bai, S., Chen, K., Liu, X., Wang, J., Ge, W., Song, S., Dang, K., Wang, P., Wang, S., Tang, J., Zhong, H., Zhu, Y., Yang, M., Li, Z., Wan, J., Wang, P., Ding, W., Fu, Z., Xu, Y., Ye, J., Zhang, X., Xie, T., Cheng, Z., Zhang, H., Yang, Z., Xu, H., Lin, J.: Qwen2.5-vl technical report. arXiv preprint arXiv:2502.13923 (2025)
- [4] Chen, X., Xu, H., Ruan, M., Bian, M., Chen, Q., Huang, Y.: So-tad: A surveillance-oriented benchmark for traffic accident detection. *Neurocomputing* **618**, 129061 (2025)
- [5] Dhuliawala, S., Komeili, M., Xu, J., Raileanu, R., Li, X., Celikyilmaz, A., Weston, J.: Chain-of-verification reduces hallucination in large language models. arXiv preprint arXiv:2309.11495 (2023)
- [6] Du, H., Nan, G., Qian, J., Wu, W., Deng, W., Mu, H., Chen, Z., Mao, P., Tao, X., Liu, J.: Exploring what why and how: A multifaceted benchmark for causation understanding of video anomaly. arXiv preprint arXiv:2412.07183 (2024)
- [7] Du, H., Zhang, S., Xie, B., Nan, G., Zhang, J., Xu, J., Liu, H., Leng, S., Liu, J., Fan, H., et al.: Uncovering what, why and how: A comprehensive benchmark for causation understanding of video anomaly. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. pp. 18793–18803 (2024)
- [8] Eikema, B., Aziz, W.: Is MAP decoding all you need? the inadequacy of the mode in neural machine translation. In: *Proceedings of the 28th International Conference on Computational Linguistics (COLING)* (2020)
- [9] Gao, S., Yang, P., Guo, H., Liu, Y., Chen, Y., Li, S., Zhu, H., Xu, J., Zhang, X.Y., Huang, L.: The evolution of video anomaly detection: A unified framework from dnn to mllm. arXiv preprint arXiv:2507.21649 (2025)
- [10] Gu, S., Wang, X., Ying, D., Zhao, H., Yang, R., Jin, M., Li, B., Pavone, M., Yeung-Levy, S., Wang, J., et al.: Accidentbench: Benchmarking multimodal

- understanding and reasoning in vehicle accidents and beyond. arXiv preprint arXiv:2509.26636 (2025)
- [11] Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W.: LoRA: Low-rank adaptation of large language models. In: International Conference on Learning Representations (ICLR) (2022)
 - [12] Huang, C., Wang, B., Wen, J., Liu, C., Wang, W., Shen, L., Cao, X.: Vad-r1: Towards video anomaly reasoning via perception-to-cognition chain-of-thought. In: Advances in Neural Information Processing Systems (NeurIPS) (2025)
 - [13] Kong, Q., Kawana, Y., Saini, R., Kumar, A., Pan, J., Gu, T., Ozao, Y., Opra, B., Sato, Y., Kobori, N.: Wts: A pedestrian-centric traffic video dataset for fine-grained spatial-temporal understanding. In: Proceedings of the European Conference on Computer Vision (ECCV) (2024)
 - [14] Kumar, S., Byrne, W.: Minimum Bayes-Risk decoding for statistical machine translation. In: Proceedings of the Human Language Technology Conference of the North American Chapter of the Association for Computational Linguistics (HLT-NAACL) (2004)
 - [15] Li, J., Li, D., Savarese, S., Hoi, S.: BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In: International Conference on Machine Learning (ICML) (2023)
 - [16] Lin, B., Ye, Y., Zhu, B., Cui, J., Ning, M., Jin, P., Yuan, L.: Video-LLaVA: Learning united visual representation by alignment before projection. In: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP). pp. 6094–6104. Association for Computational Linguistics (November 2024)
 - [17] Lin, D., Qu, M., Han, K., Jiao, J., Jin, X., Wei, Y.: A unified reasoning framework for holistic zero-shot video anomaly analysis. In: Advances in Neural Information Processing Systems (NeurIPS) (2025)
 - [18] Lv, H., Zhou, C., Cui, Z., Xu, C., Li, Y., Yang, J.: Localizing anomalies from weakly-labeled videos. *IEEE Transactions on Image Processing* **30**, 4505–4515 (2021)
 - [19] Maaz, M., Rasheed, H., Khan, S., Khan, F.S.: Video-ChatGPT: Towards detailed video understanding via large vision and language models. In: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL) (2024)
 - [20] Pezeshkpour, P., Hruschka, E.: Large language models sensitivity to the order of options in multiple-choice questions. In: Findings of the Association for Computational Linguistics: NAACL 2024. pp. 2006–2017 (2024)
 - [21] Piehl, M., Ye, M.: LATERN: Test-time context-aware explainable video anomaly detection. arXiv preprint arXiv:2605.15054 (2026)
 - [22] Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., Sutskever, I.: Learning transferable visual models from natural language supervision. In: International Conference on Machine Learning (ICML). pp. 8748–8763. Proceedings of Machine Learning Research, PMLR (2021)

- [23] Razi, A., Chen, X., Li, H., Wang, H., Russo, B., Chen, Y., Yu, H.: Deep learning serves traffic safety analysis: A forward-looking review. *IET Intelligent Transport Systems* **17**(1), 22–71 (2023)
- [24] Shah, A.: Highway traffic videos dataset. <https://www.kaggle.com/datasets/aryashah2k/highway-traffic-videos-dataset> (2020)
- [25] Sultani, W., Chen, C., Shah, M.: Real-world anomaly detection in surveillance videos. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. pp. 6479–6488 (2018)
- [26] Tang, J., Lu, H., Wu, R., Xu, X., Ma, K., Fang, C., Guo, B., Lu, J., Chen, Q., Chen, Y.C.: Hawk: Learning to understand open-world video anomalies. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2024)
- [27] Tang, Z., Wang, S., Anastasiu, D.C., Chang, M.C., et al.: The 10th AI City Challenge. In: *ECCV Workshops*. Malmö, Sweden (2026)
- [28] Tang, Z., Wang, S., Anastasiu, D.C., et al.: The 9th ai city challenge. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)* (2025)
- [29] Wang, P., Li, L., Chen, L., Cai, Z., Zhu, D., Lin, B., Cao, Y., Liu, Q., Liu, T., Sui, Z.: Large language models are not fair evaluators. In: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)* (2024)
- [30] Wang, P., Bai, S., Tan, S., Wang, S., Fan, Z., Bai, J., Chen, K., Liu, X., Wang, J., Ge, W., Fan, Y., Dang, K., Du, M., Ren, X., Men, R., Liu, D., Zhou, C., Zhou, J., Lin, J.: Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution. *arXiv preprint arXiv:2409.12191* (2024)
- [31] Wang, S., Anastasiu, D.C., Tang, Z., Chang, M.C., Yao, Y., Zheng, L., Rahman, M.S., Arya, M.S., Sharma, A., Chakraborty, P., et al.: The 8th AI City Challenge. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops* (2024)
- [32] Xu, Y., Huang, C., Nan, Y., Lian, S.: Tad: A large-scale benchmark for traffic accidents detection from video surveillance. *arXiv preprint arXiv:2209.12386* (2022)
- [33] Yu, Y., Cao, C., Wang, Z., Meng, W., Li, J., Li, Y., Wei, Z., Shen, Z., Zhang, J.: Cuebench: Advancing unified understanding of context-aware video anomalies in real-world. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. pp. 12187–12195 (2026). <https://doi.org/10.1609/aaai.v40i14.38209>
- [34] Zhang, H., Xu, X., Wang, X., Zuo, J., Huang, X., Gao, C., Zhang, S., Yu, L., Sang, N.: Holmes-VAU: Towards long-term video anomaly understanding at any granularity. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (2025)
- [35] Zheng, C., Zhou, H., Meng, F., Zhou, J., Huang, M.: Large language models are not robust multiple choice selectors. In: *The Twelfth International Conference on Learning Representations (ICLR)* (2024)
- [36] Zindi: Barbados traffic analysis challenge. <https://zindi.africa/competitions/barbados-traffic-analysis-challenge/data> (2020)